Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

WISER Global Quantum+AI Program 2026 — BQP Challenge
Quantum-Assisted Physics-Informed Neural Networks for CFD

Draft — July 2026

Abstract

Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) replace the first hidden layer of a PINN with a variational quantum circuit (VQC). Prior work reports that this maintains accuracy while reducing trainable parameters, but the *mechanism* — when, why, and how the quantum layer changes learning — is unexplained. We build a verified, parameter-matched, PDE-agnostic test bench (viscous Burgers and heat, each with an exact reference) and a five-dimension explainability (XAI) suite, and use them to ask a single sharp question: is the quantum layer's behaviour governed by the *frequency band* it adds (the Schuld–Sweke–Meyer Fourier picture), or by something else? A controlled one-knob-at-a-time ablation (depth, entanglement, post-net width) on Burgers, each paired with a parameter-matched classical control, gives three suggestive findings: (i) QAPINN accuracy and *realized bandwidth* both grow with re-upload depth, and the quantum model overtakes the matched classical net only once the band is wide enough; (ii) at *fixed* band, changing entanglement does *not* improve accuracy and slightly degrades it, contradicting the common "entanglement-driven expressibility" narrative; and (iii) against a *structurally identical* classical-front-end control (isolating quantum vs classical front-end on the same post-net), the quantum advantage is *bandwidth-gated* — it ties the control at low depth/qubits and wins only when the band is wide enough — and the apparent classical-side effects (neuron diversity, gradient amplification) do *not* survive this fair control: the diversity effect is inconsistent across knobs and the gradient amplification reverses at higher qubit count (a barren-plateau signature). We stress that all numbers are from a single seed on one, undertrained PDE and are therefore *hypotheses, not claims*; the pipeline itself is independently verified (8/8 correctness checks). The report closes with the exact statistical work required to promote these hypotheses to results.

Contents

1 Introduction **2**

2 Background and Related Work **2**

3 Methodology **3**
3.1 Test problems 3
3.2 Models 3
3.3 Explainability suite 3
3.4 Pipeline verification 4

4 Experiments and Results **4**
4.1 Baseline comparison 4
4.2 Tier-1 ablation 4
4.3 Tier-3 ablation: qubits, measurement, encoding scale 5
4.4 How the quantum layer changes the classical network 6

5 Discussion **7**

6 Limitations **8**

7 Conclusion **8**

A Reproducibility **9**

1 Introduction

Physics-Informed Neural Networks (PINNs) [1] solve ODEs/PDEs by making the differential equation itself the loss: a fully-connected network $u_\theta(x,t)$ is trained to minimise the PDE residual together with initial and boundary conditions. A Quantum-Assisted PINN (QAPINN) [4] is the hybrid variant obtained by *replacing the first hidden layer with a VQC*. The reference benchmark reports up to $\sim$20% fewer trainable parameters at maintained accuracy on 1-D Burgers.

The BQP challenge asks not to prove QAPINNs beat PINNs, but to *explain* the learning dynamics and to derive a methodology for problem-specific circuit design. This report takes the dominant theoretical lens for VQCs — that a data-encoding circuit is a truncated Fourier series [2] — and asks whether it is sufficient to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to.

Contributions.

1. A verified, parameter-matched, PDE-agnostic bench (Burgers + heat, exact references) and a five-dimension XAI suite (spectral / F-principle, neurons, VQC internals, loss landscape, generalisation).
2. An empirical confirmation, on our own implementation, that the VQC is a truncated Fourier series whose bandwidth equals the re-upload depth.
3. A Tier-1 ablation (depth, entanglement, post-net width) isolating whether the advantage is a bandwidth effect or an entanglement effect.
4. A classical-side analysis quantifying how the quantum layer changes the downstream classical network's neurons, weights, and gradient dynamics.

2 Background and Related Work

VQCs as Fourier series. Schuld, Sweke and Meyer [2] prove that a VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$, whose accessible frequency set Ω is fixed by the eigenvalues of the encoding generators, while the coefficients c_ω are set by the trainable gates. Repeating (re-uploading) the encoding enlarges Ω; with L re-uploads of a Pauli-Y rotation the accessible frequencies are the integers $\{-L,\dots,L\}$. This is the analysis language of the present work.

Frequencies are not enough. Mhiri et al. [5] refine this: the achievable *coefficients* are constrained (a coefficient's variance scales with its frequency's redundancy) and can vanish exponentially with qubit count. Having a frequency available is necessary but not sufficient to realise it.

Quantum solvers and QAPINNs. Kyriienko et al. [3] solve nonlinear DEs with differentiable quantum circuits using analytic derivatives. Hybrid QAPINN/QCPINN designs [4, 6] report parameter reduction at matched-or-better accuracy, but gains are PDE-dependent (quantum sometimes loses, e.g. on wave and cavity flow [6]), single-group, and rarely compute-matched or reproduced. A distinct trainability pathology — a "black-hole" collapse to the trivial solution — has recently been named [7]. Across the field, quantum gains are attributed to "entanglement-driven expressibility" as a heuristic, with no validated mechanism — the gap this study targets.

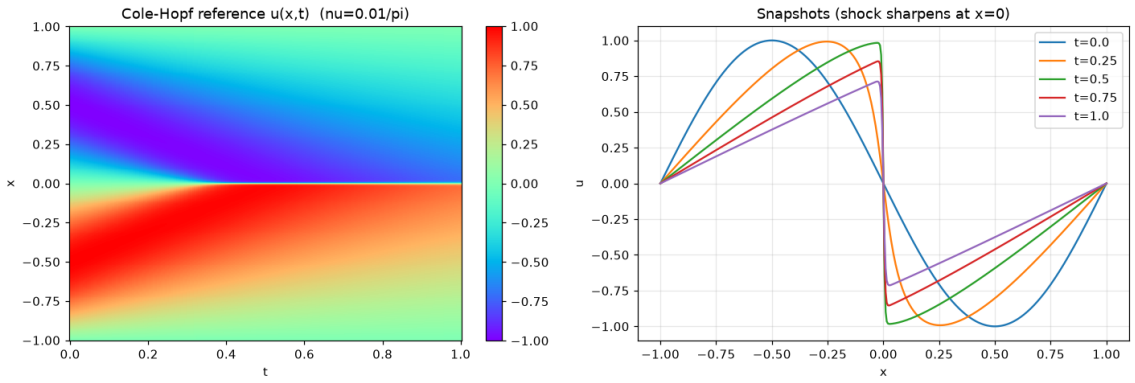


Figure 1: Exact Cole–Hopf reference for viscous Burgers. Left: field $u(x,t)$; right: snapshots showing the shock forming at $x = 0$. This is the self-generated ground truth used for all relative-L_2 scoring.

3 Methodology

3.1 Test problems

Both PDEs share the same domain $x \in [-1, 1]$, $t \in [0, 1]$, initial condition $u(x, 0) = -\sin(\pi x)$, and homogeneous Dirichlet BCs, so the *only* difference is the operator — hence the spectral content of the solution:

$$\text{Burgers:} \quad u_t + u\,u_x = \nu\,u_{xx}, \quad \nu = 0.01/\pi \qquad \text{(shock} \to \text{broadband)} \tag{1}$$

$$\text{Heat:} \quad u_t = \alpha\,u_{xx}, \quad \alpha = 1/\pi^2 \qquad \text{(single mode } k{=}1,\ u = -\sin(\pi x)e^{-t}\text{)} \tag{2}$$

Burgers uses an exact Cole–Hopf reference (Gauss–Hermite quadrature, machine precision); heat is analytic. Figure 1 shows the Burgers reference: a sine wave steepening into a shock at $x = 0$.

3.2 Models

Both models emit a scalar u and are compared at *matched trainable parameter count*.

- **Classical PINN:** a tanh MLP.
- **QAPINN** (ref. [4]): $(x, t) \to$ VQC $\to$ classical post-net $\to u$. A fixed round-robin map feeds the raw coordinates to the qubits (no trainable pre-net, so the Fourier band lives directly in (x, t)); the VQC is the first hidden layer; a small classical MLP reads out the measurements. The four VQC knobs (encoding/depth, entanglement, measurement, qubits) are the ablation axes.

The baseline VQC is 4 qubits, depth 3, RY angle encoding with re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] — 121 trainable parameters, of which only 24 are quantum.

3.3 Explainability suite

Five static dimensions plus dynamic monitoring, applied identically to both models: (1) *spectral* — realised Fourier coefficients and the F-principle (frequency learning order); (2) *neurons* — per-neuron receptive fields, dominant frequency, and effective rank; (3) *VQC internals* — per-qubit $\langle Z \rangle$ and von Neumann entanglement entropy; (4) *loss landscape*; (5) *generalisation* (extrapolation beyond the training domain). Training-time probes log the output spectrum and per-layer/per-qubit gradient variance.

Check	Method	Result
Burgers residual	vs analytic $u_t + uu_x - \nu u_{xx}$	1.2×10^{-7}
Heat residual	exact solution $\to$ residual	6.0×10^{-8}
Autograd u_x	vs finite differences	match
VQC = Fourier, max freq = depth	L=2$\to$2, L=4$\to$4	pass
relative-L_2 metric	exact solution $\to$ 0	4.0×10^{-8}
Matched parameters	all configs	121
Reproducibility	same seed $\to$ identical	bit-exact

Table 1: Correctness audit (8/8 pass). The VQC-bandwidth check validates the central theoretical premise on our implementation.

3.4 Pipeline verification

Before drawing any conclusion we independently audited every load-bearing computation (Table 1); all eight checks pass. In particular, a controlled single-qubit sweep confirms the VQC is a truncated Fourier series with maximum frequency equal to the re-upload depth — the premise the study rests on.

4 Experiments and Results

4.1 Baseline comparison

At matched 121 parameters, identical training (Adam, 2000 iters, seed 0): heat is solved well by both (classical 8.3×10^{-3}, QAPINN 5.9×10^{-3}); Burgers is undertrained for both (classical 0.433, QAPINN 0.472). The spectra (Fig. 2) tell the story: heat is a single k=1 mode captured by both, whereas the Burgers reference is broadband and *both* models undershoot the high harmonics — the depth-3 VQC is bandwidth-capped near $k \approx 3$.

4.2 Tier-1 ablation

Depth = bandwidth. Varying re-upload depth (Fig. 3a), QAPINN accuracy improves monotonically ($0.57 \to 0.49 \to 0.47 \to 0.42$) while the matched classical net stays flat ($\sim$0.50–0.53). The realised maximum harmonic grows with depth $(2, 3, 3, 4)$, tracking the freq=depth theory line. A *crossover* appears near depth 2: below it the QAPINN is comparable-or-worse than the equal-parameter classical net; at depth ≥ 3 it pulls ahead. The advantage *emerges with bandwidth*.

Entanglement (band fixed). Holding depth fixed and varying the entangling pattern (Fig. 3b), accuracy barely moves (0.456–0.488) and *more* entanglement is slightly *worse*; zero entanglement is the best. The entanglement entropy swings from 0 to $\sim$1 without tracking accuracy. This contradicts the "entanglement-driven expressibility" narrative: at fixed band, entanglement does not carry the advantage.

Table 2 lists every Tier-1 run with its parameter-matched classical control (all rel-L_2, single seed, Burgers, 2000 Adam iters).

Post-net width. As the classical post-net grows ([ ] $\to$ [8] $\to$ [16] $\to$ [32]), QAPINN accuracy improves and, for any post-net $\geq$ [8], beats the matched classical control at every width (Fig. 4).

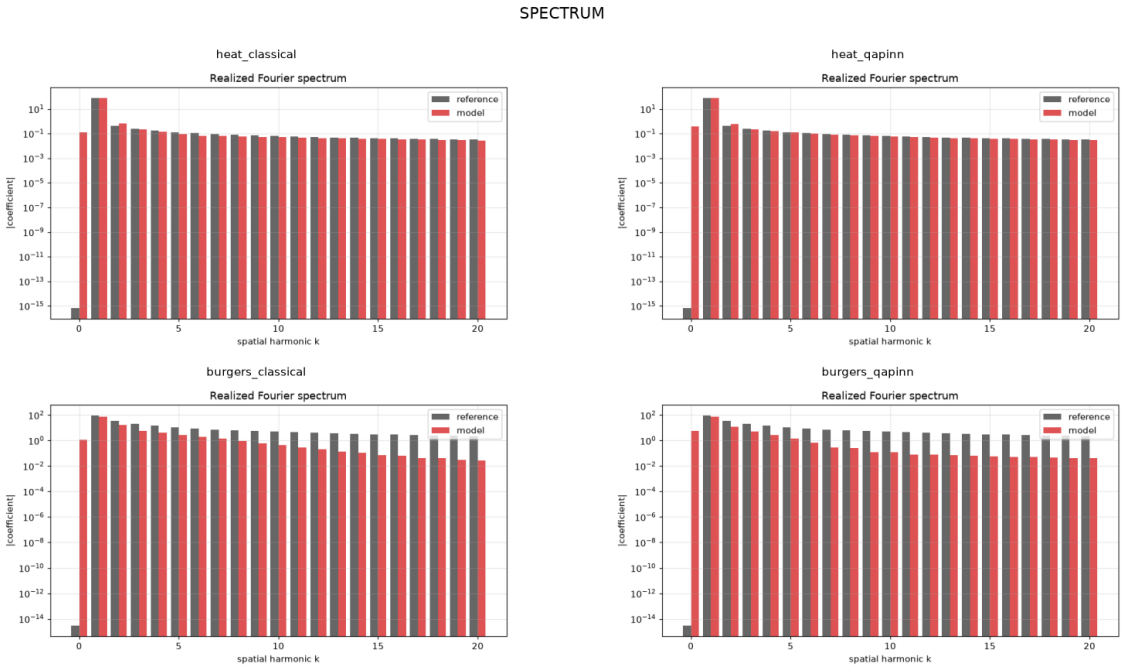


Figure 2: Realised Fourier spectra (grey: reference, red: model). Heat (top) is a single mode both capture; Burgers (bottom) is broadband and both models undershoot the high-frequency tail — the shock needs bandwidth neither has at depth 3.

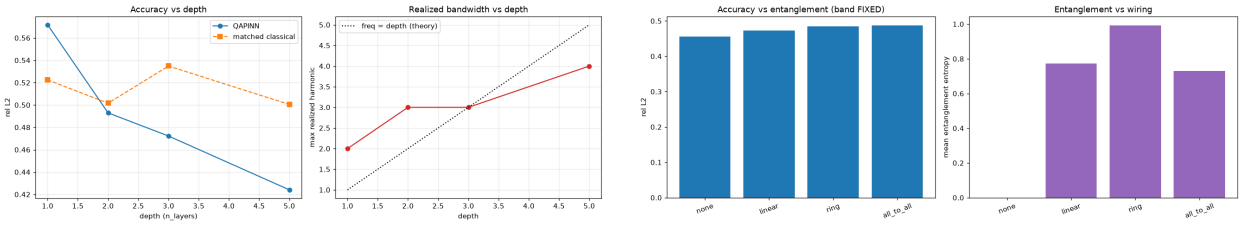
The bare-linear QAPINN is worst and loses to classical: the VQC needs a modest readout, but the classical part is not doing all the work.

Visual ablation gallery. Figures 5–7 show the ablations directly. As depth grows the QAPINN's realised spectrum (Fig. 5) captures more of the high-frequency shock tail and the solution field (Fig. 6) sharpens; across entangling patterns the entanglement entropy (Fig. 7) varies substantially while accuracy does not (Table 2).

4.3 Tier-3 ablation: qubits, measurement, encoding scale

Extending the sweep to the remaining VQC knobs (Table 3, Fig. 8) reinforces a single picture. **Qubits:** accuracy improves monotonically ($0.501 \to 0.443$) and the QAPINN overtakes the matched classical at ≥ 4 qubits — the same bandwidth crossover as depth. **Measurement:** a single $\langle Z_0 \rangle$ readout is an information bottleneck (0.591), whereas multi-Z and ZZ correlators are identical (0.472) — readout *width* matters, correlator *type* does not. **Encoding scale:** a clear optimum near 1.0, with a catastrophic *collapse to the trivial solution* at 2π (rel-$L_2 \to 1.0$) — the black-hole trainability pathology, reproduced as an ablation.

The five knobs split cleanly. *Bandwidth* knobs (depth, qubits, encoding scale) carry the advantage — but only up to a trainability ceiling, made concrete by the 2π collapse. *Coefficient/readout* knobs (entanglement, correlator measurement) do not. This is the bandwidth-vs-trainability squeeze, observed across five independent knobs, strengthening the single-seed picture beyond any one pairwise comparison.



(a) Depth sweep.

(b) Entanglement sweep (band fixed).

Figure 3: Tier-1 ablations on Burgers. (a) Accuracy and realised bandwidth grow with depth; QAPINN overtakes matched classical at depth ≥ 3. (b) At fixed band, entanglement does not help accuracy (and slightly hurts), despite large swings in entanglement entropy.

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Depth	$L = 1$	105	0.572	0.522
	$L = 2$	113	0.493	0.502
	$L = 3$ (baseline)	121	0.472	0.535
	$L = 5$	137	**0.424**	0.500
Entanglement (band fixed)	none	121	**0.456**	0.535
	linear	121	0.472	0.535
	ring	121	0.485	0.535
	all-to-all	121	0.488	0.535
Post-net	[] (linear)	29	0.544	0.533
	[8]	73	0.478	0.533
	[16] (baseline)	121	0.472	0.535
	[32]	217	**0.466**	0.505

Table 2: Complete Tier-1 ablation results. QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* is best). All single-seed.

4.4 How the quantum layer changes the classical network

The central explainability question is what the quantum front-end does to the classical post-net behind it. A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons markedly more diverse and drives larger gradients — but that comparison used a *differently shaped* classical net and is therefore confounded. We instead built a *structurally identical* classical-front-end control (CFE): the same post-net fed by a classical Linear$(2 \to w)$+tanh in place of the VQC, so the only difference is a quantum vs classical front-end. Table 4 reports both models' post-net neurons across depth, heat, and qubit count.

Against the fair control the earlier story does *not* hold. **Accuracy** is *bandwidth-gated*: the QAPINN ties the CFE at low depth/qubits and wins only at depth 5, 6 qubits, and heat. **Diversity** is *inconsistent* across knobs — the classical front-end yields more diverse post-net neurons on the depth axis, the quantum front-end more on the qubits axis — so there is no robust "quantum diversifies neurons" effect (the earlier 2× figure was an artifact of the unmatched control). **Gradient amplification** also fails to generalise: the quantum front-end drives larger post-net gradients on depth and heat, but this *reverses* at 5–6 qubits, where the QAPINN's gradient variance instead *shrinks with qubit count* — a barren-plateau signature (Fig. 9).

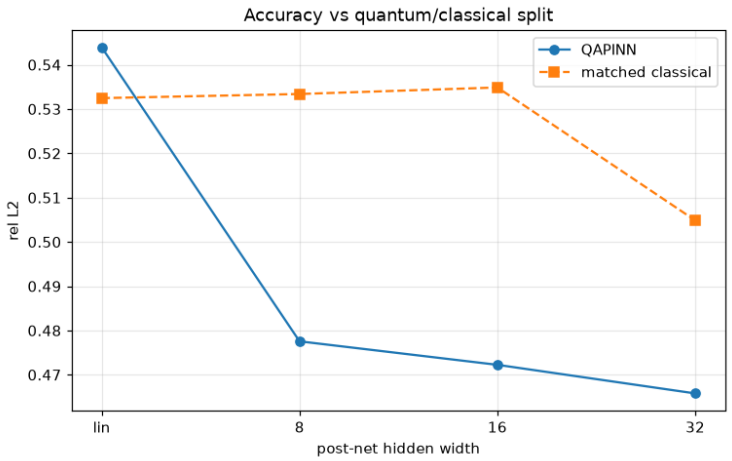


Figure 4: Post-net width sweep: QAPINN vs matched classical rel-L_2. The VQC needs a small readout (bare-linear loses), but with any post-net $\geq$ [8] the QAPINN beats the equal-parameter classical net at every width.

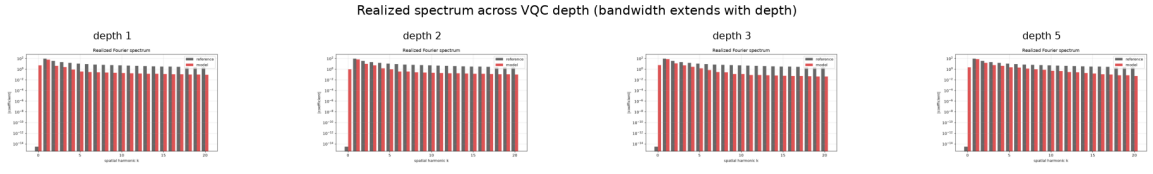


Figure 5: Realised Fourier spectrum (grey reference, red model) across VQC depth. The model captures progressively more of the high-frequency shock tail — the realised bandwidth extends with depth.

The honest conclusion: under a fair, structurally-matched control the classical-side "mechanism" does *not* yield a clean robust effect. What survives is (i) a *bandwidth-gated* accuracy advantage (the fair control ties the QAPINN at low bandwidth, and the plain-PINN comparison overstated the gap), and (ii) a *barren-plateau signature* — the QAPINN's post-net gradient variance decreases with qubit count.

5 Discussion

The evidence across all five ablated knobs is internally coherent and points one way: the quantum layer's behaviour is, so far, a *bandwidth* phenomenon. Advantage tracks the frequency-setting knobs (depth, qubits, encoding scale — each provably or empirically widening the band), and is invariant to the coefficient machinery (entanglement, correlator measurement) at fixed band; it requires — but is not subsumed by — a sufficiently wide classical readout, and holds only up to a trainability ceiling (the input_scale$= 2\pi$ collapse). Crucially, against a *structurally identical* classical-front-end control the advantage is *bandwidth-gated* (it ties at low bandwidth), and the apparent classical-side "mechanism" (neuron diversity, gradient amplification) does *not* survive the fair control — diversity is inconsistent across knobs and gradient amplification reverses at high qubit count (a barren-plateau signature). The entanglement result remains a concrete push-back against the field's default "entanglement-driven expressibility" explanation.

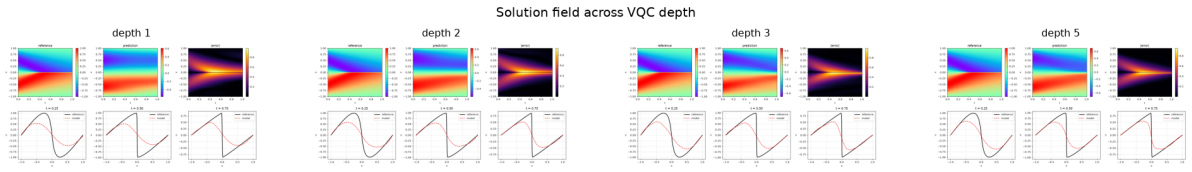


Figure 6: QAPINN solution field (reference / prediction / |error| and snapshots) across VQC depth — the shock region is fit progressively better.

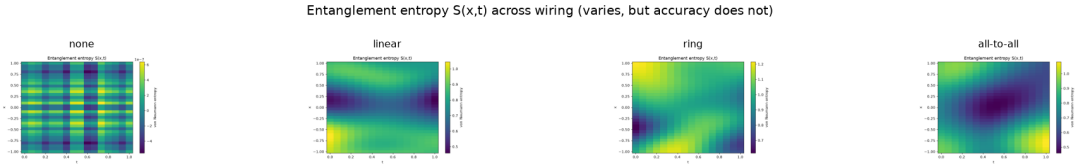


Figure 7: Entanglement entropy $S(x,t)$ across entangling patterns. Entropy varies strongly with wiring (none → ring), yet accuracy is essentially unchanged (Table 2) — entanglement is not carrying the advantage at fixed band.

6 Limitations

These are hypotheses, not claims. (1) **Single seed:** every number is one run; pairwise gaps (e.g. the entanglement spread of ∼7%) are within plausible initialisation noise, and only monotonic multi-point trends (depth, gradient-variance) carry weak additional credibility. (2) **Undertrained, one PDE:** Burgers sits at rel-L_2 0.42–0.57 under Adam-only training; the orderings could change at convergence. (3) **Classical-side effects refuted by the fair control:** the apparent diversity/gradient effects did not survive the structurally-identical CFE control, and even the surviving findings (bandwidth-gated advantage, barren-plateau signature) are single-seed. (4) **Simulation only** (`default.qubit`, statevector, no shot noise); parameter reduction is not resource reduction. The code is verified (Table 1); the *conclusions* are not yet.

7 Conclusion

Using a verified, parameter-matched bench and a five-dimension XAI suite, we find suggestive evidence that a QAPINN's behaviour is governed by the frequency band the VQC adds rather than by entanglement. Against a structurally-identical classical-front-end control the advantage is *bandwidth-gated*, and the apparent effects on the downstream classical network do not survive the fair control — what remains is a bandwidth-gated accuracy advantage and a barren-plateau signature in qubit count. Promoting these hypotheses to claims requires the multi-seed and convergence experiments detailed in the companion (extended) report.

References

[1] M. Raissi, P. Perdikaris, G. E. Karniadakis. *Physics-informed deep learning (Part I)*, 2019.

[2] M. Schuld, R. Sweke, J. J. Meyer. *Effect of data encoding on the expressive power of variational QML models*. Phys. Rev. A **103**, 032430 (2021).

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Qubits	3	99	0.501	0.470
	4 (baseline)	121	0.472	0.535
	5	143	0.468	0.541
	6	165	**0.443**	0.465
Measurement	single $\langle Z_0 \rangle$	73	0.591	0.533
	multi $\langle Z_i \rangle$ (baseline)	121	0.472	0.535
	ZZ correlators	169	0.472	0.515
input_scale	0.5	121	0.513	0.535
	1.0 (baseline)	121	**0.472**	0.535
	π	121	0.516	0.535
	2π	121	**1.000** (collapse)	0.535

Table 3: Tier-3 ablation results (single seed). Qubits behave like depth (bandwidth); readout width matters but correlator type does not; encoding scale has an optimum and collapses to the trivial solution at 2π.

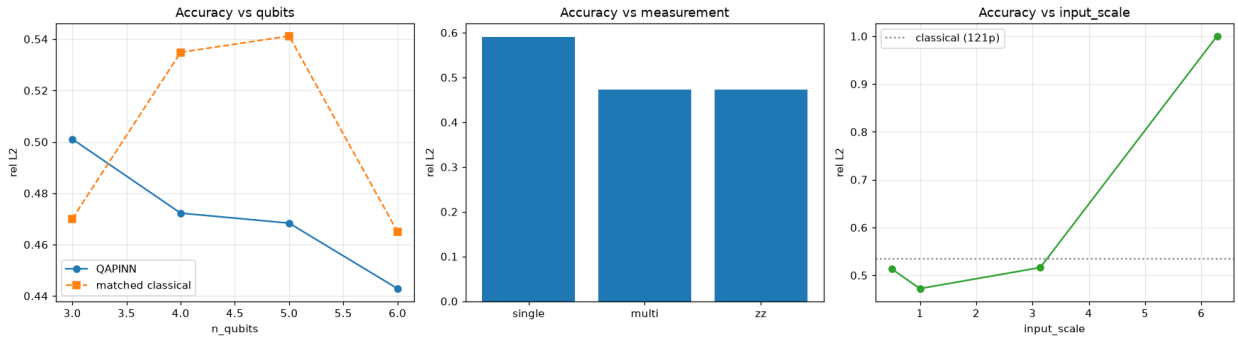


Figure 8: Tier-3 sweeps. Left: accuracy vs qubits (QAPINN overtakes classical at ≥ 4). Middle: single-Z readout bottlenecks, multi = ZZ. Right: encoding scale has an optimum near 1 and collapses at 2π.

[3] O. Kyriienko, A. E. Paine, V. E. Elfving. *Solving nonlinear DEs with differentiable quantum circuits.* Phys. Rev. A **103**, 052416 (2021).

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[5] S. Mhiri et al. *Constrained and Vanishing Expressivity of Quantum Fourier Models.* Quantum **9**, 1847 (2025).

[6] A. Farea, M. Khan. *QCPINN: Quantum-Classical PINNs for PDEs.* Mach. Learn.: Sci. Technol. (2025).

[7] Chen et al. *Quantum PINNs for Maxwell's Equations.* Quantum Machine Intelligence (2026).

A Reproducibility

Every run is driven by a YAML config and a fixed seed; the full VQC configuration is serialised to `config.json`. Runs: `python run.py experiments/<name>.yaml`; ablations: `python run_ablation.py run`; correctness audit: `python verify_pipeline.py`. Simulator: PennyLane

Comparison	rel-L_2 (Q / C)	diversity (Q / C)	grad-var $\times 10^{-3}$ (Q / C)
Burgers depth-3	0.472 / 0.474	0.269 / **0.286**	**4.1** / 0.62
Burgers depth-5	**0.424** / 0.474	0.236 / **0.286**	**7.1** / 0.62
Heat	**0.006** / 0.013	**0.272** / 0.239	**3.2** / 0.47
Burgers 3 qubits	0.501 / 0.496	**0.266** / 0.241	**1.2** / 0.26
Burgers 5 qubits	0.468 / 0.464	**0.312** / 0.265	1.1 / **5.2**
Burgers 6 qubits	**0.443** / 0.467	**0.341** / 0.269	0.72 / **1.9**

Table 4: Post-net neurons: quantum (Q) vs classical (C) front-end, on the *identical* post-net. Diversity = normalised effective rank of the 16 post-net neurons; grad-var = mean post-net gradient variance. Bold marks the larger value in each Q/C pair. Single seed.

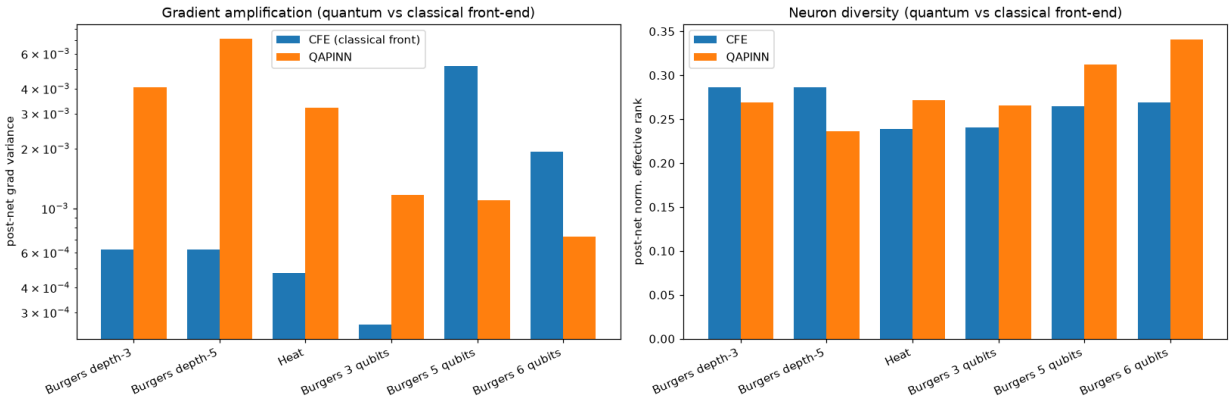


Figure 9: Quantum vs classical front-end on the identical post-net. Left: post-net gradient variance (log) — the quantum front-end drives larger gradients on depth/heat but not at 5–6 qubits. Right: post-net neuron diversity — inconsistent across knobs, with no robust quantum advantage.

`default.qubit` (statevector, backprop). See `literature_review.md`, `methodology.md`, and `xai_methods.md` for the full method notes.